

Motivation Problem Statement	Research Question	Scientific publication
- Long-distance BEV travel depends on en-route fast-charging, yet planning involves high uncertainty across decarbonization pathways (fleet size, driving range, modal shift) - The challenge of ensuring sufficient geographic coverage while avoiding overcapacity from misaligned assumptions	RQ1 What are the required fast-charging capacities along Austria's highway network for battery-electric passenger cars in 2030 under different decarbonization scenarios, and where are these allocated?	Golab, A., Zwickl-Bernhard, S., & Auer, H. (2022). Minimum-cost fast-charging infrastructure planning for electric vehicles along the Austrian high-level road network. <i>Energies</i>, 15(6), 2147. https://doi.org/10.3390/en15062147
- Broad BEV adoption requires a shift from home to public charging, while adoption needs to be considered beyond the early adopters that are characterized by high income - Cross-regional spillover effects: local charging infrastructure expansion influences BEV adoption in neighboring regions	RQ2 - To what extent does the spatial density of public charging infrastructure influence BEV adoption across consumers with different income levels? - How does a local rapid charging infrastructure expansion affect BEV adoption in neighboring regions with slower expansion?	Golab, A., Zwickl-Bernhard, S., & Auer, H. (2026). Public charger expansion: Impacts across income classes and beyond regional borders. <i>Transportation Research Part D: Transport and Environment</i>. In press.
passenger car fleets commercial fleets		
- Charging costs for battery-electric trucks vary geographically due to differences in electricity prices and network fees across countries - Unknown magnitudes of potential charging infrastructure capacity allocations being affected by these differences	RQ3 To what extent do geographically varying charging costs — arising from electricity price differentials and network charges across market zones — affect the spatial patterns of charging load distribution for international long-haul battery-electric trucks?	Golab, A., Bakker, S., Zwickl-Bernhard, S., & Auer, H. (2026). Spatial distribution of charging patterns of inter-zonal long-haul battery-electric truck fleets. <i>Energy</i>. Submitted, March 2026.
- Estimating future charging demand from multiple commercial vehicle segments (light-duty trucks, heavy-duty trucks, buses) requires segment-specific modeling approaches due to their distinct operational patterns and geographic reach - Fleet-projection-based methods miss transit freight traffic — vehicles passing through a country — which can account for a substantial share of charging demand	RQ4 What is the expected spatio-temporal distribution of charging demand from the commercial battery-electric vehicle fleet in Austria in 2040?	Golab, A., Loschan, C., Zwickl-Bernhard, S., & Auer, H. (2025). The value of flexibility of commercial electric vehicle fleets in the redispatch of congested transmission grids. <i>Energy</i>, 316, 134385. https://doi.org/10.1016/j.energy.2025.134385